

Project: MediPulse - Clinical Logistics & Inventory Intelligence Platform

1. Introduction

MediPulse is a web-based Clinical Logistics & Inventory Intelligence Platform designed for hospitals, clinics, and healthcare networks to achieve end-to-end visibility and control over clinical supplies, pharmaceuticals, medical devices, and cold-chain logistics. The platform consolidates procurement, inventory, lot/expiry tracking, replenishment, recalls, and distribution data to provide real-time stock positions, exception detection (stockouts, expiries, temperature excursions), and predictive insights for proactive clinical supply management.

MediPulse exposes RESTful APIs for backend services and supports Angular or React frontends, with reference implementations in Java (Spring Boot) and .NET (ASP.NET Core). The architecture is designed for secure handling of clinical and operational data, auditability, and integration with EHR/EMR systems via HL7/FHIR.

Actors / Users

- **Clinical Supply Manager:** Oversees hospital-wide inventory, replenishment, and expiry management.
- **Pharmacy Manager:** Manages drug inventory, lot control, expiry, and regulatory reporting.
- **Biomedical Engineer / Device Manager:** Tracks medical device inventory, maintenance schedules, and recalls.
- **Procurement Officer:** Manages supplier commitments, purchase orders, and inbound receipts.
- **Cold Chain Operator:** Monitors temperature-sensitive shipments and sensor alerts.
- **Nursing / Ward Staff:** Views local stock, requests replenishment, and records consumption.
- **Compliance Officer:** Reviews audit trails, recall actions, and regulatory exports.
- **Admin:** Configures master data, roles, thresholds, and system integrations.

2. Module Overview

2.1 Identity & Access Management

Authentication, RBAC, session controls, and audit trails with healthcare compliance.

2.2 Master Data & Location Management

Suppliers, facilities, wards, storage zones (cold/freezer/ambient), and transport lanes.

2.3 Procurement & Receiving

PO creation, ASN/GRN processing, lot and batch capture, and quality hold workflows.

2.4 Inventory, Lot & Expiry Management

On-hand inventory, lot/serial tracking, expiry alerts, FIFO/FEFO policies, and stock adjustments.

2.5 Cold Chain & Environmental Monitoring

Temperature/humidity excursion detection, sensor data ingest (optional), and corrective actions.

2.6 Distribution & Consumption Tracking

Inter-facility transfers, ward replenishment, consumption logging, and chargeback interfaces.

2.7 Exception & Recall Management

Detects stockouts, expiries, temperature excursions, and manages recall workflows.

2.8 Analytics, Forecasting & Replenishment

Demand forecasting, safety stock optimization, expiry risk scoring, and replenishment recommendations.

2.9 Notifications & Alerts

In-app alerts, escalation rules, and audit of notifications.

3. Architecture Overview

- **Frontend:** Angular or React single page application for dashboards, requisitions, and workflows.
- **Backend:** RESTful microservices (Spring Boot / ASP.NET Core) with event streaming for asynchronous processing.
- **Database:** Relational DB (Postgres/MySQL/SQL Server) for transactional data; time-series or NoSQL store (InfluxDB/Cassandra/Elastic) for environmental telemetry.
- **Cache / Ephemeral Store:** Redis for session, hot lookups, and rate limiting.
- **Integration:** HL7/FHIR adapters for EHR/EMR; SFTP/API connectors for suppliers; optional IoT gateway for sensor data.
- **Security:** API gateway, mutual TLS for partner links, encryption at rest/in transit, and HSM for key management.
- **Observability:** Centralized logging, metrics, and tracing.

4. Module Wise Design

4.1 Identity & Access Management Module

Features

- User provisioning, SSO/LDAP integration, RBAC, session policies, and immutable audit logs.

Entities

- **User**

- UserID
- Name
- Role
(SupplyManager/Pharmacy/DeviceManager/Procurement/ColdChain/Nurse/Compliance/Admin)
- Email
- Phone
- **AuditLog**
- AuditID
- UserID
- Action
- Timestamp

4.2 Master Data & Location Management Module

Features

- Maintain suppliers, facilities, storage zones, and transport lanes with storage conditions.

Entities

- **Supplier**
 - SupplierID
 - Name
 - SupplierType (Manufacturer/Distributor/3PL)
 - Status
- **Facility**
 - FacilityID
 - Name
 - Type (Hospital/Clinic/Warehouse)
 - Region
- **StorageZone**
- ZoneID
- FacilityID
- Name
- TemperatureProfile (Ambient/Refrigerated/Freezer)

- Capacity

4.3 Procurement & Receiving Module

Features

- PO lifecycle, ASN/GRN processing, quality hold, and lot capture at receipt.

Entities

- **PurchaseOrder**
 - POID
 - SupplierID
 - OrderDate
 - ExpectedDeliveryDate
 - Status
- **Receipt**
- ReceiptID
- POID
- SupplierLot
- ReceivedDate
- ReceivedBy
- QualityStatus

4.4 Inventory, Lot & Expiry Management Module

Features

- Lot/serial tracking, expiry monitoring, FEFO/FIFO policies, and stock adjustments.

Entities

- **Item**
 - ItemID
 - ItemName
 - Category (Pharma/Device/Consumable)
 - UnitOfMeasure
 - StorageRequirement
- **InventoryPosition**

- InventoryID
- FacilityID
- ZoneID
- ItemID
- LotID
- QuantityOnHand
- SafetyStock
- ExpiryDate

4.5 Cold Chain & Environmental Monitoring Module

Features

- Ingest sensor telemetry, detect excursions, correlate with shipments/storage, and trigger corrective actions.

Entities

- **SensorDevice**
 - SensorID
 - DeviceType (Temp/Humidity/GPS)
 - AssignedTo (Shipment/Zone)
 - Status
- **TelemetryRecord**
- TelemetryID
- SensorID
- Timestamp
- Temperature
- Humidity
- Location

4.6 Distribution & Consumption Tracking Module

Features

- Inter-facility transfers, ward replenishment requests, consumption logging, and chargeback.

Entities

- **TransferOrder**

- TransferID
- FromFacilityID
- ToFacilityID
- ItemID
- Quantity
- Status
- **ConsumptionRecord**
- ConsumptionID
- FacilityID
- WardID
- ItemID
- QuantityUsed
- UsedBy
- Timestamp

4.7 Exception & Recall Management Module

Features

- Detect stockouts, expiries, temperature excursions, and manage recall workflows with owner assignment and audit.

Entities

- **ExceptionEvent**
 - ExceptionID
 - Type (Expiry/Stockout/Excursion/Recall)
 - ReferenceID (Inventory/Transfer/Telemetry)
 - DetectedDate
 - Severity (Low/Medium/High)
 - Status
- **RecallAction**
- RecallID
- ExceptionID
- OwnerID
- ActionDescription

- DueDate
- Status

4.8 Analytics, Forecasting & Replenishment Module

Features

- Demand forecasting, expiry risk scoring, safety stock optimization, and replenishment recommendations.

Entities

- **Forecast**
 - ForecastID
 - ItemID
 - FacilityID
 - Period
 - ForecastQuantity
 - GeneratedDate
- **ReplenishmentPlan**
- PlanID
- ItemID
- FacilityID
- SuggestedOrderQty
- Priority

4.9 Notifications & Alerts Module

Features

- In-app alerts, escalation rules, email/webhook extensibility (future), and notification history.

Entities

- **Notification**
- NotificationID
- UserID
- Message
- Category (Exception/Expiry/Receipt/Replenishment)
- Status

- CreatedDate

5. Deployment Strategy

- **Local / Dev:** Angular/React UI, Spring Boot/.NET backend, local RDBMS and lightweight message broker for development and testing.
- **Staging:** Containerized services with synthetic telemetry and test data; model sandbox for forecasting.
- **Production:** Containerized microservices (Docker) orchestrated by Kubernetes; stateless services behind API gateway; managed relational DB with read replicas; time-series store for telemetry; secure VPC and partner connectivity for supplier integrations.
- **High Availability:** Multi-AZ deployments, auto-scaling for ingestion and analytics tiers, and automated backups for critical stores.
- **Data Residency:** Support for regional deployments to meet local healthcare data residency and compliance requirements.

6. Database Design

Tables

- **User**(UserID, Name, Role, Email, Phone)
- **AuditLog**(AuditID, UserID, Action, Timestamp)
- **Supplier**(SupplierID, Name, SupplierType, Status)
- **Facility**(FacilityID, Name, Type, Region)
- **StorageZone**(ZoneID, FacilityID, Name, TemperatureProfile, Capacity)
- **PurchaseOrder**(POID, SupplierID, OrderDate, ExpectedDeliveryDate, Status)
- **Receipt**(ReceiptID, POID, SupplierLot, ReceivedDate, ReceivedBy, QualityStatus)
- **Item**(ItemID, ItemName, Category, UnitOfMeasure, StorageRequirement)
- **InventoryPosition**(InventoryID, FacilityID, ZoneID, ItemID, LotID, QuantityOnHand, SafetyStock, ExpiryDate)
- **SensorDevice**(SensorID, DeviceType, AssignedTo, Status)
- **TelemetryRecord**(TelemetryID, SensorID, Timestamp, Temperature, Humidity, Location)
- **TransferOrder**(TransferID, FromFacilityID, ToFacilityID, ItemID, Quantity, Status)
- **ConsumptionRecord**(ConsumptionID, FacilityID, WardID, ItemID, QuantityUsed, UsedBy, Timestamp)
- **ExceptionEvent**(ExceptionID, Type, ReferenceID, DetectedDate, Severity, Status)

- **RecallAction**(RecallID, ExceptionID, OwnerID, ActionDescription, DueDate, Status)
- **Forecast**(ForecastID, ItemID, FacilityID, Period, ForecastQuantity, GeneratedDate)
- **ReplenishmentPlan**(PlanID, ItemID, FacilityID, SuggestedOrderQty, Priority)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Control Tower Dashboard:** Facility map, stock heatmap, expiry risk tiles, active exceptions, and KPI widgets (Days of Stock, Expiry Exposure, Stockouts).
- **Pharmacy Dashboard:** Lot/expiry view, incoming receipts, quality holds, and regulatory export actions.
- **Cold Chain Dashboard:** Active shipments, sensor status, excursion timeline, and corrective action buttons.
- **Ward / Requisition UI:** Local stock view, quick requisition, consumption logging, and expected replenishment ETA.
- **Device Management UI:** Device inventory, maintenance schedules, recall status, and calibration records.
- **Admin Console:** Master data management, role/permission management, integration endpoints, and system health.

8. Non-Functional Requirements

- **Performance:** Support concurrent users across a multi-facility network; ingest up to **10,000 telemetry events/sec** for medium deployments; dashboard queries return within **2s** for typical views.
- **Security & Privacy:** RBAC, SSO/LDAP, TLS for all transport, encryption at rest, PII/PHI minimization and masking, audit trails, and HIPAA/GDPR compliance controls where applicable.
- **Scalability:** Horizontal scaling for ingestion, forecasting, and analytics; multi-region deployment capability; multi-tenant support for hosted customers.
- **Availability:** Target **99.9%** uptime for core inventory and exception services; graceful degradation for non-critical analytics.
- **Maintainability:** Modular microservices, API versioning, database migration strategy, and CI/CD pipelines with canary deployments.
- **Observability:** Centralized logs, metrics (Prometheus/Grafana), distributed tracing (OpenTelemetry), and alerting for SLO breaches.
- **Data Integrity:** Lot/serial immutability for traceability, transactional guarantees for inventory movements, and reconciliation jobs.

- **Compliance & Auditability:** Retention policies, immutable audit logs, recall traceability, and exportable regulatory reports.

9. Assumptions & Constraints

- **Phase 1 scope** focuses on procurement/receiving, inventory/lot/expiry management, basic cold-chain monitoring (telemetry ingest optional via batch), distribution/consumption tracking, exception and recall workflows, and dashboards.
- **Excluded in Phase 1:** Direct integration with national drug registries requiring special legal agreements; automated clinical decisioning tied to patient care; voice/SMS verification channels (planned later).
- **Telemetry:** Real-time IoT sensor integration is optional in Phase 1 and may be implemented via an IoT gateway in Phase 2.
- **Data availability:** Historical receipt, consumption, and lot data will be provided for forecasting model training and threshold calibration.
- **Regulatory:** Local legal and compliance review required before enabling production exports for regulated reporting (e.g., controlled substances, recalls).
- **Implementable** using standard Java Spring Boot or ASP.NET Core, Angular/React, relational DB tooling, and optional time-series/NoSQL stores for telemetry.